A LIGHTWEIGHT FRAMEWORK FOR PRIVACY-PRESERVING BEHAVIORAL AUTHENTICATION

Balancing Recognition Accuracy and System Latency

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February 25, 2026

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Research Context and Motivation

- **Limits of Static MFA:** Conventional security requires repetitive, intrusive manual entry for small tasks like messaging or authorizing transactions[cite: 97, 98].
- **Transition to Behavioral Biometrics:** Keystroke and mouse dynamics offer a robust alternative by verifying users through unique biomechanical patterns[cite: 99, 100].
- **Continuous Passive Verification:** Replaces one-time logins with background monitoring to ensure the authorized user remains in control without workflow interruption[cite: 100].
- **The Privacy Conundrum:** Unlike passwords, biometric traits are inherent and immutable; a raw data breach results in permanent identity loss[cite: 112, 113, 114].
- **Data Liability:** Storing raw templates in central databases creates high-risk vulnerabilities and potential digital identity theft[cite: 115].

The Technical Dichotomy and Significance

- **The Primary Problem:** An unresolved trade-off exists between system latency, user privacy, and template revocability[cite: 131].
- **Efficiency Gap:** Cryptographic tools like Homomorphic Encryption (HE) are too computationally expensive for real-time edge monitoring[cite: 118, 125].
- **Privacy Gap:** High-accuracy Deep Learning models often require storing original behavioral features, making them prone to reverse engineering[cite: 122, 123].
- **Local Template Recoverability:** This research enables "cancelable" biometrics where compromised data is reset by regenerating a **Keyed-JL Projection Matrix** locally[cite: 121, 138].
- **Significance:** Provides a mechanism to re-issue a fresh behavioral identity on the local computer without requiring the user to change their natural habits[cite: 127, 398].

Evolution of Behavioral Authentication

Reference Study	Key Contribution to This Project	Identified Gap / Limitation
Gaines et al. (1980)	Foundational Theory: Established typing rhythms (dwell/flight time) as unique for verification.	Relied on static, fixed-text passwords; insufficient for continuous authentication.
Mondal & Bours (2017)	Multimodal Fusion: Proved combining Keystroke and Mouse dynamics significantly reduces EER.	High-dimensional vectors created by concatenation slow down real-time processing.
Kim et al. (2018)	Feature Engineering: Introduced "user-adaptive" features to improve personalized accuracy.	Lacked template protection or encryption to secure stored biometric data.
Kim et al. (2024)	Deep SVDD Validation: Outperformed traditional models (6.7% EER) via time-series to image encoding.	Restricted to mobile PINs; lacked multimodal fusion or cryptographic encryption (HE).
Kiyani et al. (2020)	Continuous Authentication: Validated RNNs for ongoing user verification beyond just login.	Did not address high latency introduced when adding encryption to continuous streams.
Rahman et al. (2021)	Scalability Metrics: Provided framework for measuring error rates as user database size increases.	Addressed scalability of accuracy but not the scalability of secure template storage.

Table: Summary of Key Related Studies

The Technical Dichotomy

- **Privacy Gaps in High-Accuracy Models:** Advanced DL frameworks (RNNs/CNNs) achieve high accuracy but often store reversible behavioral features prone to reverse engineering[cite: 104, 105].
- **Efficiency Gaps in Cryptography:** Secure tools like Homomorphic Encryption (HE) impose unaffordable computational costs and high latency, making them unsuitable for real-time edge use[cite: 106, 107, 230].
- **Lack of Integrated Revocability:** Most existing frameworks focus on accuracy but lack mechanisms for "cancelable" biometrics to reset compromised identities[cite: 108, 109, 232].
- **Unimodal Limitations:** Existing privacy transformations often ignore the compensation of entropy loss, which can be mitigated through multimodal fusion (KMT)[cite: 236, 239].
- **Defined Gap:** A lack of a unified framework providing provable non-invertibility without sacrificing sub-second latency for continuous monitoring[cite: 102, 247].

Research Aim and Technical Objectives

Main Objective

To design a lightweight and privacy-friendly framework for continuous behavioral biometric authentication that efficiently addresses the trade-off between recognition performance, system delay, and template revocability[cite: 118].

- **Privacy-Preserving Pipeline:** Develop a mechanism using Keyed-JL Projections to transform high-dimensional multimodal features into a mathematically irreversible and revocable subspace.
- **Multimodal Behavioral Fusion:** Combine independent keystroke and mouse dynamics into coordinated Keystroke and Mouse Trajectory (KMT) units to increase user profile entropy.
- **Lightweight Anomaly Detection:** Implement a server-side Deep SVDD classifier to distinguish legitimate users from impostors by learning a compact hypersphere boundary[cite: 122].
- **Trade-off Evaluation:** Quantify the impact of projection dimension k on authentication performance (EER), targeting sub-200ms latency and resistance to recovery attacks.

Theoretical and Technical Foundations

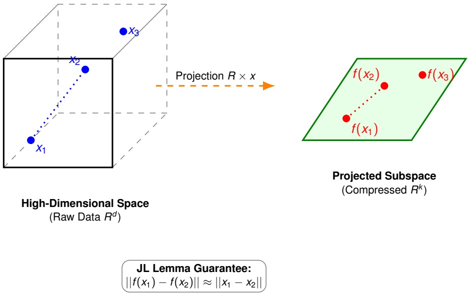


Figure: JL Projection: Dimension reduction while preserving distances[cite: 138, 147].

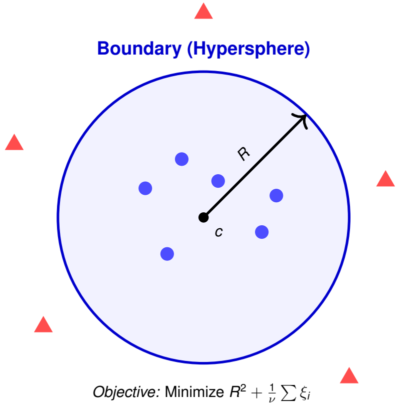
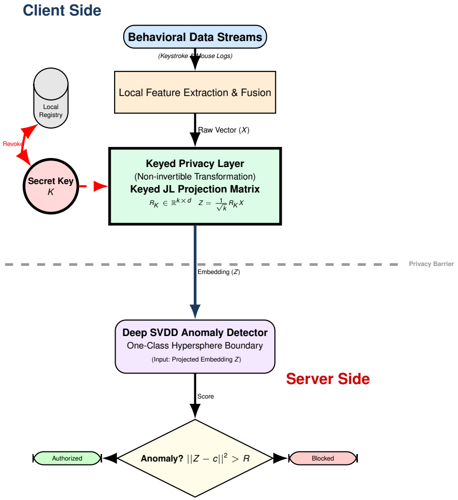


Figure: Deep SVDD: Learning a compact boundary for legitimate users[cite: 158, 170].

- **Keyed-JL:** Uses secret key for mathematical irreversibility and revocability[cite: 156].

Detailed System Architecture



- **Privacy-by-Design:** Raw data never leaves the local device, ensuring data minimization.
- **Keyed Transformation:** Uses a secret key K to generate non-invertible templates.
- **Privacy Barrier:** Only projected embeddings (Z) cross the network to the server.
- **One-Class Engine:** Deep SVDD detects anomalies without requiring imposter data.
- **Template Revocability:** Compromised identities can be "reset" by local key renewal.

Experimental Setup and Datasets

- **Privacy Protection:** Verification of mathematical non-invertibility.
- **Revocability:** Validation of local key renewal mechanisms.
- **Latency:** Target sub-200ms processing threshold.
- **Performance:** Benchmark against SU-AIS BB-MAS and CMU datasets.

Contribution to Knowledge and Novelty

- **Unified Lightweight Framework:** Introduction of a unified architecture addressing privacy, revocability, and efficient anomaly detection.
- **Integration of Random Projection and DL:** Novel combination of JL projections with Deep SVDD for secure biometric template transformation[cite: 407, 412].
- **Formal Trade-off Analysis:** Systematic evaluation of the relationship between dimensionality reduction, projection randomness, and authentication accuracy[cite: 405, 416].
- **Alternative to Heavy Cryptography:** A computationally efficient solution providing a theoretical basis for privacy without the latency of HE or SMPC[cite: 414, 415].
- **Edge-Deployable Architecture:** Advancement of research toward practical, continuous authentication for resource-constrained IoT and mobile environments[cite: 408, 417].

Assumptions and Technical Constraints

- **Modality Limitation:** The system is currently restricted to desktop inputs (KMT) and does not include mobile-specific biometrics.
- **Data Availability:** Evaluation is bounded by the diversity and volume of data present in the selected benchmark datasets (BB-MAS and CMU).
- **Latency vs. Privacy Bound:** A critical constraint exists in the trade-off between projection dimension k and system performance, targeting < 200ms.
- **Behavioral Consistency:** Results assume that the primary user's typing and mouse usage patterns remain relatively stable during the study[cite: 273].
- **Environmental Noise:** Variations in hardware peripherals (different keyboards or mice) may introduce noise affecting the Equal Error Rate (EER)[cite: 283].

Official Project Milestones (2026)

Milestone	Target Date
Research Proposal Defense	February 25, 2026
Submission of Introduction & Literature Review	May 17, 2026
Mid Review Presentation	May 27, 2026
Submission of Methodology Chapter	August 9, 2026
Submission of Full Project Report	September 27, 2026
Final Presentation	October 19–20, 2026

Table: Official Project Milestones [cite: 436]

- **Current Phase:** Literature Review and Problem Refinement (Jan–May)[cite: 439].
- **Upcoming:** System Design and Multimodal Feature Engineering (May–July)[cite: 447, 448].

Summary and Final Remarks

- **Addressing the Dichotomy:** This research bridges the gap between high-latency cryptographic privacy and privacy-invasive raw data models[cite: 111, 119].
- **Lightweight Security:** By integrating Keyed-JL Projections with Deep SVDD, the framework achieves mathematically guaranteed privacy without heavy computational costs[cite: 121, 137].
- **Local Identity Control:** The system empowers users with "cancelable" biometrics, allowing them to reset compromised identities locally by renewing projection keys[cite: 138, 384].
- **Real-Time Viability:** The proposed architecture targets sub-200ms latency, making it ideal for continuous monitoring on resource-constrained edge devices[cite: 134, 385].
- **Final Deliverable:** The project will culminate in a fully validated, open-source framework and a comprehensive performance evaluation report[cite: 421, 424].

Thank You